

The direct criterion of Newcomb for the ideal MHD stability of an axisymmetric toroidal plasma

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A method is presented for determining the ideal magnetohydrodynamic stability of an axisymmetric toroidal plasma, based on a toroidal generalization of the method developed by Newcomb for fixed-boundary modes in a cylindrical plasma. For toroidal mode number $n \neq 0$, the stability problem is reduced to the numerical integration of a high-order complex system of ordinary differential equations, the Euler-Lagrange equation for extremizing the potential energy, for the coupled amplitudes of poloidal harmonics m as a function of the radial coordinate ψ in a straight-fieldline flux coordinate system. Unlike the cylindrical case, different poloidal harmonics couple to each other, which introduces coupling between adjacent singular intervals. A boundary condition is used at each singular surface, where $m = nq$ and $q(\psi)$ is the safety factor, to cross the singular surface and continue the solutions beyond it. Fixed-boundary instability is indicated by the vanishing of a real determinant of a Hermitian complex matrix constructed from the fundamental matrix of solutions, the generalization of Newcomb's crossing criterion. In the absence of fixed-boundary instabilities, an $M \times M$ plasma response matrix $\mathbf{W}_P$, with M the number of poloidal harmonics used, is constructed from the Euler-Lagrange solutions at the plasma-vacuum boundary. This is added to a vacuum response matrix $\mathbf{W}_V$ to form a total response matrix $\mathbf{W}_T$. The existence of negative eigenvalues of $\mathbf{W}_T$ indicates the presence of free-boundary instabilities. The method is implemented in the fast and accurate DCON code. *Published by AIP Publishing.*

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I. INTRODUCTION

A key step in understanding the behavior of tokamaks is the determination of the stability of the confined plasma to ideal magnetohydrodynamic (MHD) perturbations. Such instabilities are believed to set limits on the achievable current density and plasma pressure and to influence the onset of disruptions which can terminate the discharge and damage the device. Stability studies constitute a major effort in the interpretation of experiments and design studies for new tokamaks.

Most existing MHD stability codes^{1–6} are based on a procedure in which the perturbed potential and kinetic energies are expanded in a set of two-dimensional basis functions, reducing the numerical problem to the solution of a large matrix eigenvalue problem for the growth rates, frequencies, and eigenfunctions of the entire MHD spectrum. These methods can suffer from poor convergence of the basis functions, especially for difficult equilibria with high β (the ratio $2\mu_0 P/B^2$ of plasma pressure to magnetic field energy density), strong plasma shaping, and poloidal divertors. Coupling to the singular eigenfunctions of the ideal MHD continuous spectrum introduces uncertainty into the location of the stability boundary. These methods require large expenditures of human and computer time. Much of the effort required by such methods goes into determining eigenvalues and eigenvectors which may not be needed when the real concern is simply the location of the stability boundary

and perhaps a measure of the distance of a given equilibrium from that boundary.

An alternative procedure was introduced by Newcomb for cylindrical plasmas.⁷ He found conditions for the existence of test functions which satisfy the boundary conditions and make the potential energy δW negative, relying on the well-known result that the existence of such test functions implies the existence of exponentially growing perturbations.⁸ His condition involves integrating a real 2nd-order ordinary differential equation (ODE), the Euler-Lagrange equation for minimizing the potential energy, and determining whether the solution, the radial displacement ξ , changes sign between successive singular points. This condition can be interpreted as determining whether there is a mode which fits into the “box” of the equilibrium. We refer to it as the Direct Criterion of Newcomb, or DCON.

The purpose of this paper is to generalize Newcomb's criterion to an axisymmetric toroidal plasma for perturbations with toroidal mode number $n \neq 0$. The key difference between a cylinder and a torus is that θ , the poloidal angle which goes around the short way, is no longer a symmetry in a toroidal plasma, as it is in a cylinder. As a result, one can no longer treat individual Fourier components independently in this variable, because toroidicity and noncircularity couple different components and thus require that they be treated simultaneously. The generalization of Newcomb's 2nd-order real Euler-Lagrange equation is a complex system of ODEs of order $2M$, where M is the number of poloidal Fourier components retained in the treatment. This system is solved numerically with a highly efficient adaptive ODE solver.⁹

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In order to generalize Newcomb's sign-change criterion, it is necessary to identify a real scalar quantity whose sign change plays the same role in the toroidal case as that of Newcomb's radial displacement ξ in the cylindrical case. This quantity has been found as a critical determinant constructed from the Euler-Lagrange solutions with regular initial conditions at the magnetic axis. Its reality is a consequence of the Hamiltonian symmetry of the Euler-Lagrange equation, which is in turn a consequence of the well-known self-adjointness of the linearized ideal MHD equations. Its role in the theory is demonstrated by a generalization of Newcomb's proofs to axisymmetric toroidal geometry. Another important difference between the cylindrical and toroidal cases is that in the former, successive subintervals between singular surfaces can be treated independently, while in the latter they are coupled by nonresonant components which must be C^1 continuous across singular surfaces.

Bineau¹⁰ attempted a treatment similar to ours but for general, non-axisymmetric toroidal plasmas. His approach has many of the same features as this one, but it is more formal, it is not clear how to apply it as a practical stability program, and he does not obtain our key result, the Critical Determinant. Section VII discusses the difficulties of extending our treatment to nonaxisymmetric systems. Connor *et al.*¹¹ and Fitzpatrick *et al.*¹² treat the equivalent of our Euler-Lagrange equation in the context of a large aspect ratio expansion. Their interest is in toroidal resistive tearing modes; they do not discuss the symmetries of the equations or the generalization of Newcomb's criterion for ideal stability. Dewar and Pletzer^{13,14} discuss a generalization of Newcomb's Euler-Lagrange equation to axisymmetric toroidal plasmas, but they do not express its properties as a coupled system of ODEs for the complex Fourier amplitudes, nor do they generalize Newcomb's criterion. Their concern is with resistive stability, and their numerical treatment uses an expansion in linear finite elements in the radial (ψ) direction.

The remainder of this paper is organized as follows. In Section II, we introduce a static equilibrium and coordinate system and a vector representation used to evaluate and minimize the potential energy δW , derive the Euler-Lagrange equation, and discuss the symmetries of this equation and its solutions. In Section III, we discuss the behavior of the Euler-Lagrange equation in the neighborhood of its singular points at the resonant surfaces, the magnetic axis, and the separatrix. In Section IV, we prove the toroidal generalization of Newcomb's criterion for fixed-boundary modes in terms of a critical determinant. In Section V, we extend the treatment to include a vacuum region surrounding the plasma region, enabling us to treat free-boundary instabilities. In Section VI, we discuss numerical methods and results. In Section VII, we discuss our conclusions. Appendix gives detailed expressions for the coefficient matrices which appear in the Euler-Lagrange equation. SI units are used throughout.

II. THE EULER-LAGRANGE EQUATION

In this section, we first describe the equilibrium and introduce a coordinate system and a representation for

perturbed vectors. We then use these to express the ideal MHD potential energy δW , and derive the Euler-Lagrange equation as the condition that δW be stationary with respect to small changes in the perturbation. We conclude with a discussion of the symmetry properties of this equation.

We treat a stationary equilibrium¹⁵ satisfying the pressure balance equation

$$\mathbf{J} \times \mathbf{B} = \mu_0 \nabla P, \quad (1)$$

where $\mathbf{B}$ is the magnetic field, $\mathbf{J} = \nabla \times \mathbf{B} / \mu_0$ is the plasma current, $\mathbf{B}$ and $\mathbf{J}$ are divergenceless, and P is the plasma pressure. We assume that the system is axisymmetric, i.e., all equilibrium scalars are independent of the azimuthal angle ϕ . With these assumptions, the magnetic field can be represented as

$$\mathbf{B} = f \nabla \phi + \frac{\chi'}{2\pi} \nabla \phi \times \nabla \psi, \quad (2)$$

where $f = RB_T$, R is the radius from the axis of symmetry; B_T is the toroidal component of the magnetic field; χ is the poloidal magnetic flux; P , χ , and f are constant on a flux surface, i.e., a surface of constant ψ ; ψ an arbitrary surface label, and primes denote derivatives with respect to ψ . The poloidal flux satisfies the Grad-Shafranov equation

$$\Delta^* \chi \equiv R^2 \nabla \cdot \left(\frac{1}{R^2} \nabla \chi \right) = -\frac{4\pi^2}{\chi'} (ff' + \mu_0 R^2 P'). \quad (3)$$

This static equilibrium does not allow for strong plasma rotation, velocities of order the sound or Alfvén velocities, for which inertial and anisotropic pressure effects can be important. Such fast rotation can occur in the presence of strong neutral beam injection. However, we do not exclude the case of diamagnetic rotation in which these effects are small and affect primarily the neighborhood of resonant surfaces.

Location within the region of closed magnetic field lines is described in terms of straight-fieldline coordinates. Each flux surface is labeled by an arbitrary coordinate ψ , which increases monotonically from the axis to the plasma edge. It is commonly chosen to be linear in the poloidal flux and normalized to go from 0 at the axis to 1 at the edge. Position on a flux surface is described in terms of the periodic variables θ and ζ , which increase by 1 (not 2π) after one turn around the torus the short way and the long way, respectively. Physical scalars must be single-valued, returning to their original values after θ or ζ increases by one. The system has Jacobian

$$\mathcal{J} \equiv (\nabla \psi \cdot \nabla \theta \times \nabla \zeta)^{-1}. \quad (4)$$

The contravariant representation of the equilibrium magnetic field is given by

$$\mathbf{B} = \chi' (\nabla \zeta - q \nabla \theta) \times \nabla \psi, \quad (5)$$

where q , the safety factor or winding number, is a function of ψ only. For any function $\mathcal{F}(\psi, \theta, \zeta)$, derivatives along the

magnetic field, which play a key role in MHD stability theory, have the simple representation

$$\mathbf{B} \cdot \nabla \mathcal{F} = \frac{\chi'}{\mathcal{J}} \left(\frac{\partial}{\partial \theta} + q \frac{\partial}{\partial \zeta} \right) \mathcal{F}(\psi, \theta, \zeta). \quad (6)$$

The poloidal and toroidal currents constant on a flux surface are given by

$$\mathbf{J} \cdot \nabla \theta = -\frac{2\pi f'}{\mathcal{J}}, \quad \mathbf{J} \cdot \nabla \zeta = q \mathbf{J} \cdot \nabla \theta - \frac{\mu_0 P'}{\chi'}, \quad (7)$$

while the normal current $\mathbf{J} \cdot \nabla \psi$ vanishes.

The perturbed displacement vector ξ is described in terms of its contravariant representation

$$\xi = \mathcal{J}(\xi_\psi \nabla \theta \times \nabla \zeta + \xi_\theta \nabla \zeta \times \nabla \psi + \xi_\zeta \nabla \psi \times \nabla \theta), \quad (8)$$

and a useful combination of these components, the surface displacement

$$\xi_s \equiv \mathcal{J}(\xi \times \mathbf{B}) \cdot (\nabla \theta \times \nabla \zeta) = \chi'(q\xi_\theta - \xi_\zeta). \quad (9)$$

Likewise, the perturbed magnetic field is expressed in terms of its contravariant components

$$\begin{aligned} \mathbf{Q} &= \nabla \times (\xi \times \mathbf{B}) \\ &= \mathcal{J}(Q_\psi \nabla \theta \times \nabla \zeta + Q_\theta \nabla \zeta \times \nabla \psi \\ &\quad + Q_\zeta \nabla \psi \times \nabla \theta). \end{aligned} \quad (10)$$

These components are related to the components of ξ by

$$\begin{aligned} Q_\psi &= \frac{\chi'}{\mathcal{J}} \left(\frac{\partial}{\partial \theta} + q \frac{\partial}{\partial \zeta} \right) \xi_\psi, \\ Q_\theta &= -\frac{1}{\mathcal{J}} \frac{\partial}{\partial \psi} (\chi' \xi_\psi) + \frac{1}{\mathcal{J}} \frac{\partial \xi_s}{\partial \zeta}, \\ Q_\zeta &= -\frac{1}{\mathcal{J}} \frac{\partial}{\partial \psi} (q \chi' \xi_\psi) - \frac{1}{\mathcal{J}} \frac{\partial \xi_s}{\partial \theta}. \end{aligned} \quad (11)$$

The ideal MHD perturbed potential energy is given by⁸

$$\begin{aligned} \delta W &= \frac{1}{2\mu_0} \int_{\Omega} d\mathbf{x} [Q^2 + \mathbf{J} \cdot \xi \times \mathbf{Q} \\ &\quad + \mu_0 (\xi \cdot \nabla P)(\nabla \cdot \xi) + \mu_0 \gamma P (\nabla \cdot \xi)^2], \end{aligned} \quad (12)$$

where Ω denotes the volume of the plasma region. When Eqs. (8)–(11) are substituted into Eq. (12), we obtain

$$\begin{aligned} \delta W &= \frac{1}{2\mu_0} \int_{\Omega} d\psi d\theta d\zeta \mathcal{J} \left\{ [\nabla \xi_s \times \nabla \psi + (\nabla \theta \times \nabla \zeta) \mathcal{J} \mathbf{B} \cdot \nabla \xi_\psi - \mathbf{B} \xi_\psi' - [\chi'' \nabla \zeta \times \nabla \psi + (q\chi')' \nabla \psi \times \nabla \theta] \xi_\psi]^2 \right. \\ &\quad + \xi_\psi \mathbf{J} \cdot \nabla \xi_s - \xi_s \mathbf{J} \cdot \nabla \xi_\psi + [J_\theta (q\chi')' - J_\zeta \chi''] \xi_\psi^2 + \frac{\mu_0 P'}{\mathcal{J}} (\mathcal{J} \xi_\psi^2)' \\ &\quad \left. + \frac{1}{\mathcal{J}} \frac{\partial}{\partial \theta} (\mathcal{J} \mu_0 P' \xi_\psi \xi_\theta) + \frac{1}{\mathcal{J}} \frac{\partial}{\partial \zeta} (\mathcal{J} \mu_0 P' \xi_\psi \xi_\zeta) + \gamma \mu_0 P (\nabla \cdot \xi)^2 \right\}. \end{aligned} \quad (13)$$

The first two terms on the last line are perfect derivatives which vanish on integration over a flux surface. The last term is positive-definite and is the only term containing ξ_θ . Minimization with respect to ξ_θ eliminates this term for toroidal mode number $n \neq 0$. The minimizing perturbations are thus divergenceless. This will be assumed henceforth, and the compression term will be dropped.

The remaining terms contain only the normal displacement ξ_ψ and the surface displacement ξ_s . While ξ_ψ enters through its radial derivative ξ_ψ' , as well as angular derivatives with respect to θ and ζ , ξ_s enters only through its angular derivatives, which will permit us to eliminate it from the treatment and reduce δW to a form involving only the radial displacement and its radial derivative, as in Newcomb's treatment of the cylindrical plasma.

We now introduce Fourier series for the normal and surface displacements

$$\begin{aligned} \begin{pmatrix} \xi_s \\ \xi_\psi \end{pmatrix} (\psi, \theta, \zeta) &= \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \begin{pmatrix} \bar{\xi}_s \\ \bar{\xi}_\psi \end{pmatrix} \bigg|_{m,n} (\psi) \\ &\quad \times \exp[2\pi i(m\theta - n\zeta)]. \end{aligned} \quad (14)$$

In practice, the infinite sums in Eq. (14) are truncated to include a finite number M of components. The reality of the

physical perturbations implies that the Fourier coefficients satisfy the reality conditions

$$\begin{pmatrix} \xi_s \\ \xi_\psi \end{pmatrix} = \begin{pmatrix} \xi_s \\ \xi_\psi \end{pmatrix}^* \Rightarrow \begin{pmatrix} \bar{\xi}_s \\ \bar{\xi}_\psi \end{pmatrix} \bigg|_{-m,-n} = \begin{pmatrix} \bar{\xi}_s \\ \bar{\xi}_\psi \end{pmatrix}^* \bigg|_{m,n}. \quad (15)$$

A useful, compact notation is to introduce the complex M -vectors of Fourier components of the radial and normal displacements

$$\begin{pmatrix} \Xi_s \\ \Xi_\psi \end{pmatrix} (\psi) \equiv \left\{ \begin{pmatrix} \bar{\xi}_s \\ \bar{\xi}_\psi \end{pmatrix} \bigg|_{m,n} (\psi), m_{\text{low}} \leq m \leq m_{\text{high}} \right\}. \quad (16)$$

With these definitions, the potential energy can be expressed as

$$\begin{aligned} \delta W &= \frac{1}{2\mu_0} \int_a^b d\psi [\Xi_s^\dagger \mathbf{A} \Xi_s + \Xi_s^\dagger (\mathbf{B} \Xi_\psi' + \mathbf{C} \Xi_\psi) \\ &\quad + (\Xi_\psi^\dagger \mathbf{B}^\dagger + \Xi_\psi^\dagger \mathbf{C}^\dagger) \Xi_s + \Xi_\psi^\dagger \mathbf{D} \Xi_\psi' \\ &\quad + \Xi_\psi^\dagger \mathbf{E} \Xi_\psi + \Xi_\psi^\dagger \mathbf{E}' \Xi_\psi' + \Xi_\psi^\dagger \mathbf{H} \Xi_\psi], \end{aligned} \quad (17)$$

where a and b are the values of ψ bounding the plasma region, $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{H}$ are complex $M \times M$ matrices, dagger denotes Hermitian conjugate, and $\mathbf{A}$, $\mathbf{D}$, and $\mathbf{H}$ are self-adjoint. Detailed expressions for these matrices are given in Eq. (A6) of Appendix. Since Ξ_s enters Eq. (17) only algebraically, and since $\mathbf{A}$ is found to be everywhere nonsingular, δW can be minimized with respect to Ξ simply by solving the matrix equation

$$\begin{aligned} \mathbf{A}\Xi_s + \mathbf{B}\Xi'_\psi + \mathbf{C}\Xi_\psi &= 0, \\ \Xi_s &= -\mathbf{A}^{-1}(\mathbf{B}\Xi'_\psi + \mathbf{C}\Xi_\psi). \end{aligned} \quad (18)$$

When this expression is substituted for Ξ_s in Eq. (17), we obtain a form of δW involving only Ξ_ψ and its radial derivative Ξ'_ψ only. Hereafter, we suppress the subscript on Ξ_ψ and obtain

$$\delta W = \frac{1}{2\mu_0} \int_a^b d\psi [\Xi'^\dagger \mathbf{F} \Xi' + \Xi'^\dagger \mathbf{K} \Xi + \Xi^\dagger \mathbf{K}^\dagger \Xi' + \Xi^\dagger \mathbf{G} \Xi], \quad (19)$$

in terms of the composite matrices

$$\begin{aligned} \mathbf{F} &\equiv \mathbf{D} - \mathbf{B}^\dagger \mathbf{A}^{-1} \mathbf{B} = \mathbf{F}^\dagger, \\ \mathbf{K} &\equiv \mathbf{E} - \mathbf{B}^\dagger \mathbf{A}^{-1} \mathbf{C} \neq \mathbf{K}^\dagger, \\ \mathbf{G} &\equiv \mathbf{H} - \mathbf{C}^\dagger \mathbf{A}^{-1} \mathbf{C} = \mathbf{G}^\dagger. \end{aligned} \quad (20)$$

Simplified expressions for $\mathbf{F}$ and $\mathbf{K}$ are given in Eqs. (A7) and (A8) of Appendix.

The expression in Eq. (19) is a real functional of the complex vectors Ξ and $\Xi^\dagger$. The operations of complex conjugate and Hermitian conjugate are non-analytic because they fail to satisfy the Cauchy-Riemann conditions. Because of this, Ξ and $\Xi^\dagger$ must be treated as independent vectors in any complex analytical operation, then set to their conjugate values at the end of the operation. The Euler-Lagrange equation is the condition that δW be stationary with respect to small changes in Ξ and $\Xi^\dagger$

$$-(\mathbf{F}\Xi' + \mathbf{K}\Xi)' + (\mathbf{K}^\dagger \Xi' + \mathbf{G}\Xi) = 0, \quad (21)$$

a system of M coupled 2nd-order ODEs, and its Hermitian conjugate. This is a necessary but not sufficient condition that δW be a minimum. Whether the stationary point is a minimum depends on additional considerations discussed in Section IV. However, these conditions are expressed in terms of the properties of the solutions to Eq. (21), which therefore plays a central role in the theory.

The relationship to Newcomb's Euler-Lagrange equation is now straightforward. For the cylindrical case, the Fourier components decouple. The matrices $\mathbf{F}$, $\mathbf{G}$, and $\mathbf{K}$ become diagonal and can be regarded as real scalars. In this case, the term $(\mathbf{K}\Xi)'$ can be expanded, the contribution from Ξ' cancels the other term in the equation, and the contribution from $\mathbf{K}'$ can be combined with $\mathbf{G}$ to give Newcomb's equation. In the more general toroidal case, $\mathbf{K}$ is not self-adjoint, and no real simplification is obtained by this procedure.

It is useful, for both analytical and numerical purposes, to express Eq. (21) as an equivalent system of $2M$ coupled

1st-order ODEs. This can be done by introducing the complex $2M$ -vector

$$\mathbf{u} \equiv \begin{pmatrix} \Xi \\ \mathbf{F}\Xi' + \mathbf{K}\Xi \end{pmatrix}, \quad (22)$$

and the complex $2M \times 2M$ matrix

$$\mathbf{L} \equiv \begin{pmatrix} -\mathbf{F}^{-1}\mathbf{K} & \mathbf{F}^{-1} \\ \mathbf{G} - \mathbf{K}^\dagger \mathbf{F}^{-1} \mathbf{K} & \mathbf{K}^\dagger \mathbf{F}^{-1} \end{pmatrix}, \quad (23)$$

in terms of which Eq. (21) can be written

$$\mathbf{u}' = \mathbf{L}\mathbf{u}. \quad (24)$$

The matrix $\mathbf{L}$ has an important symmetry: the off-diagonal blocks are self-adjoint, while the diagonal blocks are the negative of each other's adjoint. More formally, in terms the unit symplectic matrix

$$\mathbf{J} \equiv \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{pmatrix}, \quad (25)$$

$\mathbf{L}$ has the Hamiltonian property, $\mathbf{J}\mathbf{L}\mathbf{J} = \mathbf{L}^\dagger$. In Hamiltonian terms, the second M components of $\mathbf{u}$ in Eq. (22) can be regarded as the conjugate momenta of the first M components, the displacements. This is understood in abstract terms, since the independent variable ψ is not time, and the variables are complex Fourier components. The Hamiltonian symmetry of $\mathbf{L}$ may be thought of as a manifestation in this representation of the well-known self-adjointness of ideal MHD.⁸

Corresponding to the Hamiltonian symmetry of $\mathbf{L}$, the fundamental matrix of solutions has the property of being symplectic. Equation (24) is a $2M$ th-order ODE and therefore has $2M$ independent solutions, corresponding to different initial conditions. These $2M$ column vectors can be assembled into a $2M \times 2M$ matrix $\mathbf{U}$ which also satisfies Eq. (24), with $\mathbf{u}$ replaced by $\mathbf{U}$. $\mathbf{U}$ is a linear canonical transformation, the propagator of the system. Using this equation and its Hermitian conjugate, it is simple to show that $\mathbf{U}^\dagger \mathbf{J} \mathbf{U}$ is a conserved quantity; in fact, it is the Hamiltonian of the system. If $\mathbf{U}$ is chosen initially to be the identity, then it everywhere satisfies the equation

$$\mathbf{U}^\dagger \mathbf{J} \mathbf{U} = \mathbf{J}. \quad (26)$$

We can express $\mathbf{U}$ in terms of $M \times M$ blocks

$$\mathbf{U} = \begin{pmatrix} \mathbf{U}_{11} & \mathbf{U}_{12} \\ \mathbf{U}_{21} & \mathbf{U}_{22} \end{pmatrix}, \quad (27)$$

in which the upper blocks represent displacements, the lower blocks represent their conjugate momenta, and the left and right halves represent different sets of initial conditions. In terms of these blocks, the symplectic property is equivalent to

$$\begin{aligned} \mathbf{U}_{11}^\dagger \mathbf{U}_{21} - \mathbf{U}_{21}^\dagger \mathbf{U}_{11} &= 0, \\ \mathbf{U}_{12}^\dagger \mathbf{U}_{22} - \mathbf{U}_{22}^\dagger \mathbf{U}_{12} &= 0, \\ \mathbf{U}_{11}^\dagger \mathbf{U}_{22} - \mathbf{U}_{21}^\dagger \mathbf{U}_{12} &= \mathbf{I}. \end{aligned} \quad (28)$$

Similarly, one can show that if Eq. (26) is satisfied initially, then it retains this property, and from this we obtain the additional independent relations, so the properties in Eq. (28) are preserved. These symmetries play a central role in proving the theorems of Section IV.

III. SINGULAR SURFACES

The Euler-Lagrange equation has regular singular points at the resonant surfaces where $q = m/n$, at the magnetic axis, and at a separatrix if one exists. In this section, we discuss the asymptotic behavior of the solutions in the neighborhood of each of these singular points.

The form of the Euler-Lagrange equation, Eqs. (22)–(24), shows that the coefficient matrix $\mathbf{L}$ becomes singular whenever the determinant of $\mathbf{F}$ vanishes. Using the forms of these matrices given in Eqs. (A7) and (A8) of Appendix, we can write

$$\mathbf{F} = \mathbf{Q}\bar{\mathbf{F}}\mathbf{Q}, \quad \mathbf{K} = \mathbf{Q}\bar{\mathbf{K}}, \quad \mathbf{G} = \bar{\mathbf{G}}, \quad (29)$$

where $\bar{\mathbf{F}}$, $\bar{\mathbf{G}}$, and $\bar{\mathbf{K}}$ are nonsingular at the resonant surfaces. Since $\mathbf{Q}$ becomes singular at the resonant surfaces where $q = m/n$, these are singular points. For an axisymmetric torus, as treated here, the toroidal mode number n is a good quantum number, i.e., a fixed integer for each perturbation treated, and therefore, the resonant surfaces are discrete. In the presence of a magnetic separatrix, where $q \rightarrow \infty$ logarithmically as a function of ψ , there is a point of accumulation of the resonant surfaces. While we have not been able to prove it analytically, we find numerically that $\bar{\mathbf{F}}$ vanishes linearly with ψ and $\bar{\mathbf{G}}$ varies as ψ^{-1} as $\psi \rightarrow 0$; this corresponds to the behavior of Newcomb's equation in a cylinder, making the axis a singular point. In addition, we find numerically that $\det \bar{\mathbf{F}}$ vanishes at a separatrix, making it another singular point. In certain degenerate cases, e.g., if $q = m/n$ at the axis or at an extremum where q' vanishes, two or more singular points can merge to form a singularity of higher rank. We treat only the relatively simple cases for which this does not occur.

In analyzing the asymptotic behavior of the solution in the neighborhood of a resonant surface, we must take account of the full matrix nature of the equations and the coupling between resonant and nonresonant components due to toroidal curvature and poloidal shaping. To accomplish this, we use a matrix form of Frobenius analysis due to Turrington.¹⁶

Before treating the full case, we illustrate the procedure for Newcomb's cylindrical case, for which the Euler-Lagrange equation is

$$(f\xi')' - g\xi = 0, \quad (30)$$

where primes denote derivatives with respect to r . To express this in matrix form, let

$$\mathbf{u} = \begin{pmatrix} \xi \\ f\xi' \end{pmatrix}, \quad \mathbf{L} = \begin{pmatrix} 0 & 1/f \\ g & 0 \end{pmatrix}. \quad (31)$$

Then, Eq. (30) becomes $\mathbf{u}' = \mathbf{L}\mathbf{u}$. Let r_s be the position of the singular surface, let $z \equiv r - r_s$ for $r > r_s$, and reinterpret

the primes to denote derivatives with respect to z . The functions $f(r)$ and $g(r)$ can be Taylor expanded about the singular surface as $f = f_0 z^2 + \dots$, $g = g_0 + \dots$. Then to lowest order

$$\mathbf{L} = \begin{pmatrix} 0 & 1/f_0 z^2 \\ g_0 & 0 \end{pmatrix} + \dots \quad (32)$$

We now define a new dependent variable vector $\mathbf{v}$ by introducing the shearing transformation

$$\mathbf{u} = \mathbf{R}\mathbf{v}, \quad \mathbf{R} \equiv \begin{pmatrix} z^{-1/2} & 0 \\ 0 & z^{1/2} \end{pmatrix}. \quad (33)$$

Then, $\mathbf{v}$ satisfies the equation $z\mathbf{v}' = \mathbf{M}\mathbf{v}$ in terms of the transformed matrix

$$\mathbf{M} = z\mathbf{R}^{-1}(\mathbf{L}\mathbf{R} - \mathbf{R}') = \begin{pmatrix} 1/2 & 1/f_0 \\ g_0 & -1/2 \end{pmatrix} + \dots \quad (34)$$

If we now seek power-like solutions of the form $\mathbf{v} = z^\alpha \mathbf{v}^{(0)} + \dots$, then to lowest order we obtain the matrix eigenvalue equation

$$(\mathbf{M} - \alpha \mathbf{I})\mathbf{v} = 0, \quad (35)$$

with eigenvalues satisfying the characteristic equation

$$\det(\mathbf{M} - \alpha \mathbf{I}) = 0, \quad \alpha_\pm = \pm \sqrt{-D_I}, \quad (36)$$

$$D_I = -g_0/f_0 - 1/4, \quad (37)$$

and eigenvectors

$$\mathbf{v}_\pm = \begin{pmatrix} 1 \\ (\alpha_\pm - 1/2)f_0 \end{pmatrix}. \quad (38)$$

Equation (35) thus plays the role of the indicial equation in the standard Frobenius analysis and also determines the form of the solution vectors. Transforming these solutions back to the original vector of dependent variables $\mathbf{u}$ by means of Eq. (33), we find that the general solution can be expressed as a linear combination of large and small solutions

$$\mathbf{u} = c_- z^{\alpha_-} \begin{pmatrix} z^{-1/2} \\ z^{1/2}[\alpha_- - 1/2]f_0 \end{pmatrix} + c_+ z^{\alpha_+} \begin{pmatrix} z^{-1/2} \\ z^{1/2}[\alpha_+ - 1/2]f_0 \end{pmatrix}. \quad (39)$$

The shearing transformation introduced in Eq. (33) is required in order to properly balance the components of the new dependent variable vector $\mathbf{v}$, giving both components the same power-like behavior. If we applied the same shearing transformation at an ordinary point of the equation, or to a nonresonant component of the solution at a resonant surface, then since the components of $\mathbf{u}$ are already in balance, both approaching a constant, the shearing transformation would unbalance them. Thus, in treating the toroidal case, we must choose a shearing transformation which appropriately balances the resonant components without unbalancing

the nonresonant components. This introduces half-integer powers into the coupling between resonant and nonresonant components.

In the neighborhood of a resonant surface of a toroidal plasma, let $q(\psi_R) = m/n$, $z \equiv \psi - \psi_R$. Then, the dominant terms in $\mathbf{L}$, Eq. (23), vary as

$$\begin{aligned}\mathbf{F}^{-1} &\sim z^{-2}, \\ \mathbf{F}^{-1}\mathbf{K} &\sim \mathbf{K}^\dagger\mathbf{F}^{-1} \sim z^{-1}, \\ \mathbf{G} &\sim \mathbf{K}^\dagger\mathbf{F}^{-1}\mathbf{K} \sim 1.\end{aligned}\quad (40)$$

To balance the equations, we introduce the shearing transformation

$$\begin{aligned}\mathbf{u} &= \mathbf{R}\mathbf{v}, \quad R_{m,m'} = z^{r_m}\delta_{m,m'}, \\ r_m &= -1/2, \quad r_{m+M} = 1/2 \quad \text{for } m = nq, \\ r_m &= r_{m+M} = 0 \quad \text{for } m \neq nq,\end{aligned}\quad (41)$$

where M is the number of retained Fourier components. Then, the equation is transformed to

$$z\mathbf{v}' = \mathbf{M}\mathbf{v}. \quad (42)$$

The transformed matrix

$$\mathbf{M} = z\mathbf{R}^{-1}(\mathbf{L}\mathbf{R} - \mathbf{R}') \quad (43)$$

can be Taylor expanded about the resonant point as

$$\mathbf{M} = \sum_{k=0}^{\infty} z^{k/2} \mathbf{M}_k. \quad (44)$$

We now seek power-like solutions of the form

$$\mathbf{v} = z^\alpha \sum_{k=0}^{\infty} z^{k/2} \mathbf{v}^{(k)}. \quad (45)$$

Introducing Eqs. (44) and (45) into Eq. (42), we obtain, as the lowest-order terms, the matrix eigenvalue equation, $[\mathbf{M}_0 - \alpha\mathbf{I}]\mathbf{v}^{(0)} = 0$. The eigenvalues α are the solutions to the characteristic equation

$$\det[\mathbf{M}_0 - \alpha\mathbf{I}] = 0. \quad (46)$$

Higher-order terms in the power series solutions can be obtained iteratively from the higher-order equations

$$[\mathbf{M}_0 - (\alpha + k/2)\mathbf{I}]\mathbf{v}^{(k)} = -\sum_{l=1}^k \mathbf{M}_l \mathbf{v}^{(k-l)}. \quad (47)$$

Since this is a regular singular point, the powers series solutions are convergent.

For the degenerate cases where α is a half integer, Eq. (47) is singular at some value of k . Among these is the case of vanishing plasma pressure, for which $\alpha = \pm 1/2$. We have worked out full power series solutions for this case, characterized by terms containing $\ln z$ as well as just powers of z , but it is more complicated and beyond the scope of this paper.

The spectrum of Eq. (46) is simple. Since $\mathbf{M}_0$ is a $2M \times 2M$ matrix, there are $2M$ eigenvalues and eigenvectors. Of these, two represent the resonant solutions corresponding to Eq. (37) for the cylinder. The other $2M - 2$ eigenvalues are zero, corresponding to nonresonant solutions.

Determination of the resonant eigenvalues can be reduced to the solution of a 2×2 characteristic equation. We take two basis vectors, each containing 1 for one of its resonant components and all other components zero. Multiplication of these vectors by $\mathbf{M}_0$ annihilates all contributions from the null space. The resonant components of the resulting vectors form a 2×2 matrix whose eigenvalues are

$$\alpha = \pm \sqrt{-D_I}, \quad (48)$$

where D_I is the well-known expression for the Mercier criterion in axisymmetric toroidal geometry¹⁷⁻¹⁹

$$\begin{aligned}D_I &= -\frac{1}{4} + \frac{2\pi f P' V'}{q' \chi'^3} \left\langle \frac{1}{|\nabla \psi|^2} \right\rangle \left[1 - \frac{2\pi f P' V'}{q' \chi'^3} \left\langle \frac{1}{|\nabla \psi|^2} \right\rangle \right] \\ &\quad + \left\langle \frac{B^2}{|\nabla \psi|^2} \right\rangle \frac{P' V'^2}{q'^2 \chi'^4} \left\{ P' \left[\left\langle \frac{1}{B^2} \right\rangle + \left(\frac{2\pi f}{\chi'} \right)^2 \left\langle \frac{1}{B^2 |\nabla \psi|^2} \right\rangle \right] + \frac{\chi''}{\chi'} - \frac{V''}{V'} \right\},\end{aligned}\quad (49)$$

where V is the volume inclosed within a flux surface. While we have not been able to prove analytically that the solutions to Eq. (46) are in fact given by Eqs. (48) and (49), we find numerically that they converge to these values as the number of retained Fourier components increases. Agreement between these two ways of computing D_I can thus serve as a test for adequate Fourier convergence at a resonant surface. The Mercier criterion, that $D_I \leq 0$ is a necessary condition for stability, follows from that fact that if $D_I > 0$, then α is imaginary, the solution vector $\mathbf{v}$ in Eq. (45) is oscillatory,

and the generalization of Newcomb's stability criterion, presented in Sec. IV, is repeatedly violated.

Each solution vector $\mathbf{v}$ varies as a single power of z , either α , Eq. (48), for the resonant solutions, or 0 for the nonresonant solutions. Because of the half-integer powers in the shearing transformation, the resonant displacements of the resonant vectors $\mathbf{u}$ contains an extra power $1/\sqrt{z}$ and the resonant conjugate momenta of the resonant vectors $\mathbf{u}$ contain an extra power of $\sqrt{z}$ relative to the nonresonant components. The iterative solution procedure Eq. (47) couples

integer resonant components to half-integer nonresonant components, and vice versa. The power series can be determined to arbitrarily high order.

Once all of the independent eigenvectors have been determined to some order, they can be assembled as the columns of a matrix which we denote $\mathbf{V}(\psi)$. We choose a particular order for these vectors, as close as possible to the identity. The subspace of nonresonant eigenvectors, corresponding to the eigenvalues $\alpha = 0$, is spanned by a basis in which each basis vector contains only one nonresonant component to lowest order. We arrange these vectors so that the nonvanishing component appears on the diagonal and normalize them so that the diagonal element is 1. We place the large resonant eigenvector in the resonant column of the left half of $\mathbf{V}$ and the small resonant eigenvector in the resonant column of the right half of $\mathbf{V}$. With a proper choice of normalization of the resonant solutions, $\mathbf{V}$ can be made symplectic.

In principle, the convergent power series could be evaluated to all orders, and $\mathbf{V}$ would then be an exact fundamental solution matrix. Any particular solution vector $\mathbf{u}(\psi)$ of the exact Euler-Lagrange equation, such as one obtained by integrating the equations numerically, could then be expressed as a linear combination of these independent solutions, $\mathbf{u}(\psi) = \mathbf{V}(\psi)\mathbf{c}$, where the constant vector $\mathbf{c}$ consists of the expansion coefficients in a linear combination. In practice, it is often more convenient to truncate the power series after a few terms, in which case $\mathbf{V}(\psi)$ is not an exact fundamental solution matrix. We can still express an exact solution as a linear combination of the truncated power series solutions

$$\mathbf{u}(\psi) = \mathbf{V}(\psi)\mathbf{c}_A(\psi). \quad (50)$$

Then, the vector $\mathbf{c}_A(\psi)$ of expansion coefficients is no longer constant, but asymptotically approaches a limit as the resonant surface is approached, as long as the number of terms in the power series is sufficiently large. We therefore denote the components of this vector asymptotic coefficients. As $\psi \rightarrow \psi_R$, $\mathbf{c}_A$ can be obtained by solving Eq. (50).

The two resonant eigenvectors corresponding to the eigenvalues in Eq. (48) have powers differing by $2\sqrt{-D_I}$. If accurate determination of the asymptotic coefficient of the small resonant solution is necessary, as in the treatment of singular modes, then enough terms in the series solution of the large solution must be retained so that its remaining error is smaller than the zeroth-order term in the small solution. If $|D_I|$ is large, this can make the procedure numerically difficult. This typically occurs in regions of very low shear q' , for example, when q has an extremum; or when β , the ratio of plasma pressure to magnetic energy density is large enough to make terms proportional to P' in D_I , Eq. (49), large compared to unity.

The axis is a singular point of the Euler-Lagrange equation, just as it is for Newcomb's cylindrical case. For the latter, the solutions for $r\xi_r$ vary as $r^{\pm m}$ in the neighborhood of the axis. Boundary conditions at the axis must exclude the r^{-m} solutions and retain the r^m solutions. Similarly, one would expect that, in the toroidal case, solutions for $\xi \cdot \nabla \psi$

should vary as $\psi^{\pm\mu/2}$, with μ an integer. Determination of the asymptotic behavior at the axis should be reducible to an analysis similar to that presented above for the resonant surfaces, based on a matrix eigenvalue equation of the form

$$\lim_{\psi \rightarrow 0} (\psi \mathbf{L} - \alpha \mathbf{I}) \mathbf{u} = 0. \quad (51)$$

There are several complications which make this difficult. The complex form of $\mathbf{F}$ given in Eq. (A7), especially coupling to the $m = 0$ term, makes it difficult to extract the asymptotic behavior of $\mathbf{L}$ analytically. Since the eigenvalues differ by integers, there are further complications, as discussed by Turrittin.

Equation (51) can be treated numerically as the solution to a matrix eigenvalue problem. Doing so, we find that the eigenvalues do indeed approach $\mu/2$, as expected. The eigenvectors obtained by this means are used to initialize the integration near the axis. In general, a pure eigenvector, corresponding to pure $\psi^{\mu/2}$ behavior, contains a mixture of all Fourier components m , due to toroidal (± 1) and elliptic (± 2) coupling in the neighborhood of the axis.

If a separatrix bounds the region of confined plasma, it has two effects on the distribution of singular surfaces. First, it causes $q \rightarrow \infty$, introducing an accumulation point of singular surfaces due to the $\mathbf{Q}$ factors in $\mathbf{F}$, Eq. (A7). Second, we have found numerically that the remaining factors of $\mathbf{F}$ have a vanishing determinant at the separatrix. If we treat a truncated Fourier series with some maximum value of m , then this additional singularity can be regarded to play a role similar to that of the singular point at the axis. A full understanding of this behavior remains to be achieved.

At each resonant point, one eigenvalue of $\mathbf{F}$ has a quadratic zero; it does not change sign. The only other singularities of $\mathbf{F}$ are at the axis and the separatrix. Thus, throughout the region of closed field lines between the axis and the separatrix, all eigenvalues of $\mathbf{F}$ are non-negative. This property is used in Sec. IV to generalize the proofs of the stability theorems.

IV. FIXED-BOUNDARY MODES

Our goal in this paper is to determine whether there exist test functions $\Xi(\psi)$ which make δW negative while satisfying the boundary conditions. In our analysis so far, we have derived the Euler-Lagrange equation which makes δW stationary with respect to variations of Ξ , and we have studied its symmetry properties and its singular points. A necessary condition for an extremum of δW is that the test function satisfy the Euler-Lagrange equation.^{20,21} Now we must find a sufficient condition for the stationary solution to be a minimum. In this section, we study the case of fixed-boundary modes, with $\Xi = 0$ at the plasma-vacuum interface, $\psi = 1$, the only case treated by Newcomb. In Sec. V, we consider free-boundary modes, in which the interface is allowed to move, $\Xi(1) \neq 0$, and there is a perturbed magnetic field in the vacuum beyond the plasma.

In Newcomb's treatment of the cylindrical screw pinch, the symmetry with respect to cylindrical coordinates θ and z allows these variables to be eliminated in terms of single Fourier components (m, n). The potential energy δW (denoted W by Newcomb), Eq. (15), is expressed as a quadratic functional of the real scalar quantity $\xi(r)$, the radial plasma displacement, and its first derivative $\xi'(r)$ with respect to r , and he proves that the intervals between singular surfaces may be treated independently. This is a problem in the calculus of variations, based on Ref. 20. The central concept of his proofs is the Hilbert Invariant Integral, as discussed in Chapter V of Ref. 20.

As a result of generalization to axisymmetric toroidal geometry, symmetry with respect to the poloidal coordinate θ is broken. This requires that the single Fourier component m be extended to a set of M coupled Fourier components $m \in [m_{\text{low}}, m_{\text{high}}]$, whose amplitudes are complex. Thus, the real scalar quantity $\xi(r)$ is generalized to a complex vector $\Xi(\psi)$ of complex Fourier amplitudes, and δW becomes a real functional of a complex vector. Only one component of Ξ is resonant at each singular surface. The nonresonant components couple adjacent intervals between singular surfaces, preventing their independent treatment.

In the more recent and complete Ref. 21, two separate treatments are given of sufficient conditions for an extremum, one for a weak extremum and the other for a strong extremum, both for the case of multiple real dependent variables. The difference is in the admissible space of functions: a weak extremum is one for which the function and its first derivative are both assumed to have finite L^2 norms, while a strong extremum is one for which only the function values have finite L^2 norm. Of the two separate treatments, only the strong case uses the Hilbert Invariant Integral, as does Newcomb. Neither of these spaces is appropriate for our case, nor does this reference treat the case of a real functional of a complex vector or multiple singular surfaces.

We define a new space of admissible functions by the condition that the potential energy itself be finite, $|\delta W| < \infty$, as in Ref. 14. This requires that only regular solutions be allowed at the magnetic axis and that in the neighborhood of each singular surface, the large resonant solution be excluded and the nonresonant solutions be C^1 continuous. A procedure for numerically constructing elements of this space is given in Section VI. We follow the proof for the weak extremum, Sections 26 and 29 of Ref. 21, generalizing it to our case.

For the fixed-boundary case, using Eq. (19), the plasma potential energy can be written

$$\delta W = \frac{1}{2\mu_0} \int_0^1 d\psi [\Xi'^\dagger \mathbf{F} \Xi' + \Xi'^\dagger \mathbf{K} \Xi + \Xi^\dagger \mathbf{K}^\dagger \Xi' + \Xi^\dagger \mathbf{G} \Xi], \quad (52)$$

where $\psi = 0$ denotes the magnetic axis and $\psi = 1$ denotes the plasma-vacuum interface. For boundary conditions, we assume Ξ is bounded at $\psi = 0$, excluding the irregular solutions at the magnetic axis, and $\Xi = 0$ at $\psi = 1$, for fixed-boundary modes.

We seek conditions for δW to be positive-definite. For any Hermitian matrix $\mathbf{W}$, the integral of the perfect derivative $(\Xi^\dagger \mathbf{W} \Xi)'$ vanishes

$$\begin{aligned} \frac{1}{2\mu_0} \int_0^1 (\Xi^\dagger \mathbf{W} \Xi)' d\psi \\ = \frac{1}{2\mu_0} \int_0^1 (\Xi'^\dagger \mathbf{W} \Xi + \Xi^\dagger \mathbf{W} \Xi' + \Xi^\dagger \mathbf{W}' \Xi) d\psi = 0. \end{aligned} \quad (53)$$

We can therefore subtract Eq. (53) from Eq. (52)

$$\begin{aligned} \delta W = \frac{1}{2\mu_0} \int_0^1 [\Xi'^\dagger \mathbf{F} \Xi' + \Xi'^\dagger (\mathbf{K} - \mathbf{W}) \Xi \\ + \Xi^\dagger (\mathbf{K}^\dagger - \mathbf{W}) \Xi' + \Xi^\dagger (\mathbf{G} - \mathbf{W}') \Xi] d\psi, \end{aligned} \quad (54)$$

without changing its value. A proper choice of $\mathbf{W}$ enables us to complete the square in the integrand and verify that $\delta W \geq 0$. We choose

$$\mathbf{W} \equiv \mathbf{U}_{22} \mathbf{U}_{12}^{-1}, \quad (55)$$

where $\mathbf{U}$ satisfies the Euler-Lagrange equation and initial conditions

$$\mathbf{U}' = \mathbf{L} \mathbf{U}, \quad \mathbf{U}_{21}(0) = 0, \quad \mathbf{U}_{22}(0) = \mathbf{I}, \quad (56)$$

with the first subscript 2 of the $M \times M$ block matrices denoting the solutions which are regular at the magnetic axis. As shown in Eq. (28), it follows from the Hamiltonian symmetry of $\mathbf{L}$, the consequent symplectic symmetric of $\mathbf{U}$, and the initial conditions that $\mathbf{W}$ is Hermitian. We further define the critical determinant

$$D_C \equiv \det \mathbf{W}^{-1}. \quad (57)$$

Since $\mathbf{W}$ is Hermitian, D_C is real. It vanishes at $\psi = 0$. Any other point $0 < \psi \leq 1$ for which D_C vanishes is called a conjugate point. At such a point, $\mathbf{W}$ has a non-integrable singularity and is therefore not a member of the admissible space of functions. The condition that $\mathbf{W}$ exists and is continuous for $0 < \psi \leq 1$ is that there are no conjugate points. This is the appropriate generalization of Newcomb's crossing criterion.

From the Euler-Lagrange equation, we obtain

$$\begin{aligned} \mathbf{W}' &= \mathbf{U}_{22}' \mathbf{U}_{12}^{-1} - \mathbf{U}_{22} \mathbf{U}_{12}^{-1} \mathbf{U}_{12}' \mathbf{U}_{12}^{-1} \\ &= [(\mathbf{G} - \mathbf{K}^\dagger \mathbf{F}^{-1} \mathbf{K}) \mathbf{U}_{12} + \mathbf{K}^\dagger \mathbf{F}^{-1} \mathbf{U}_{22}] \mathbf{U}_{12}^{-1} \\ &\quad - \mathbf{U}_{22} \mathbf{U}_{12}^{-1} \mathbf{F}^{-1} (\mathbf{U}_{22} - \mathbf{K} \mathbf{U}_{12}) \mathbf{U}_{12}^{-1} \\ &= \mathbf{G} - \mathbf{K}^\dagger \mathbf{F}^{-1} \mathbf{K} - \mathbf{W} \mathbf{F}^{-1} \mathbf{W} \\ &\quad + \mathbf{K}^\dagger \mathbf{F}^{-1} \mathbf{W} + \mathbf{W} \mathbf{F}^{-1} \mathbf{K}. \end{aligned} \quad (58)$$

Substituting Eqs. (55) and (58) in Eq. (54), we obtain

$$\begin{aligned} \delta W &= \frac{1}{2\mu_0} \int_0^1 [\Xi'^\dagger + \Xi^\dagger (\mathbf{K}^\dagger - \mathbf{W}) \mathbf{F}^{-1}] \\ &\quad \times \mathbf{F} [\Xi' + \mathbf{F}^{-1} (\mathbf{K} - \mathbf{W}) \Xi] d\psi. \end{aligned} \quad (59)$$

This is a real functional of the complex vector Ξ and its Hermitian conjugate and their derivatives, with $\mathbf{F}$ a Hermitian-positive-definite matrix, except at the singular points where

$m=nq$ and its determinant vanishes, as discussed in the [Appendix](#). The integrand and hence the integral are therefore real-positive-definite, assuming that there are no conjugate points for $0 < \psi \leq 1$. The plasma is then stable to fixed-boundary modes.

This proves that the absence of conjugate points is a sufficient condition for δW to be positive-definite. We now prove that it is also a necessary condition, or equivalently, that if δW is positive-definite for all Ξ , then there are no conjugate points.

We first note the lemma that if Ξ satisfies the Euler-Lagrange equation, Eq. (21), and the boundary conditions $\Xi(0) = \Xi(1) = 0$, then δW , Eq. (52), vanishes. This follows from multiplying Eq. (21) by $\Xi^\dagger$ and integrating by parts.

Next, assuming $\delta W \geq 0$, we construct the positive-definite functional

$$\mathcal{L}(t) = \frac{1}{2\mu_0} \int_0^1 \left\{ (1-t) \Xi'^\dagger \Xi' + t [\Xi'^\dagger (\mathbf{F}\Xi' + \mathbf{K}\Xi) + \Xi^\dagger (\mathbf{K}^\dagger \Xi' + \mathbf{G}\Xi)] \right\} d\psi, \quad (60)$$

with Euler-Lagrange equation

$$-(1-t)\Xi'' + t[-(\mathbf{F}\Xi' + \mathbf{K}\Xi) + (\mathbf{K}^\dagger \Xi' + \mathbf{G}\Xi)] = 0, \quad (61)$$

which reduces to Eq. (21) for $t = 1$ and to $\Xi'' = 0$ for $t = 0$. The solution $\Xi(t)$ to Eq. (61) is a continuous function of t .

We now prove by contradiction that if δW is positive-definite, then there can be no conjugate points, $D_C = \det \mathbf{W} = 0$ for $0 < \psi \leq 1$. Suppose there exists such a point $\tilde{\psi}$. Then, there exists a nontrivial solution $\Xi(\psi)$ which vanishes at $\tilde{\psi}$, and there exists a solution $\Xi(\psi, t)$ of Eq. (61) which depends continuously on t and reduces to $\Xi(\psi)$ for $t = 0$. This cannot occur for $\tilde{\psi} = 1$, because then the positive-definite δW would vanish for $\Xi \neq 0$. Any point $0 < \tilde{\psi} < 1$ is excluded for the reasons given on page 110 of Ref. 21.

Thus, we have proven the following theorem, which generalizes Newcomb's results for the cylindrical case.

Theorem: The absence of a conjugate point, where D_C vanishes for $0 < \psi < 1$, is both necessary and sufficient for δW to be positive-definite and for the plasma to be stable to fixed-boundary modes.

V. FREE-BOUNDARY MODES

Section IV describes a procedure for determining the stability of the plasma to fixed-boundary modes, for which the normal displacement vector Ξ vanishes at the edge of the plasma. The plasma potential energy δW is positive-definite if and only if the critical determinant D_C , Eq. (57), has no zeroes for $0 < \psi \leq 1$, otherwise δW can be negative and the plasma is unstable to fixed-boundary modes. This is the only case treated by Newcomb, although he extends the treatment to include a perfectly conducting pressureless plasma with infinite conductivity, but not a vacuum region outside the plasma, with zero conductivity. The purpose of this section is to extend the analysis to include free-boundary modes, in which the plasma region is bounded by a vacuum region rather than a conducting wall.

For free-boundary modes, the normal displacement vector Ξ is finite at the plasma-vacuum interface. Motion of the interface perturbs the magnetic field in the vacuum, which introduces a positive-definite and therefore stabilizing vacuum energy contribution. Solution for the vacuum energy is performed by the VACUUM code^{22,23} by solving a Fredholm integral equation around the plasma-vacuum boundary and all of the external conductors, using a collocation method to obtain a Green's function. Using the Green's function, it then computes a vacuum response matrix $\mathbf{W}_V$ in a Fourier basis, using any specified poloidal coordinate θ . This is added to a plasma response matrix to obtain a total response matrix

$$\mathbf{W}_T = \mathbf{W}_P + \mathbf{W}_V, \quad (62)$$

whose eigenvalues and eigenvectors are computed. The plasma is unstable to free-boundary modes if and only if $\mathbf{W}_T$ has negative eigenvalues. The eigenvectors can be used to visualize the boundary motion, as shown in Fig. 4 of Ref. 24.

To compute the plasma response matrix $\mathbf{W}_P$, we start with Eq. (19) for δW , where the displacement vector Ξ satisfies the Euler-Lagrange equation, Eq. (21), and its Hermitian conjugate. We symmetrize δW by taking half the sum of the integrand and its Hermitian conjugate

$$\delta W = \frac{1}{4\mu_0} \int_0^1 d\psi [\Xi'^\dagger (\mathbf{F}\Xi' + \mathbf{K}\Xi) + \Xi^\dagger (\mathbf{K}^\dagger \Xi' + \mathbf{G}\Xi) + (\Xi'^\dagger \mathbf{F} + \Xi^\dagger \mathbf{K}^\dagger) \Xi' + (\Xi'^\dagger \mathbf{K} + \Xi^\dagger \mathbf{G}) \Xi]. \quad (63)$$

Next, we use the Euler-Lagrange equation and its Hermitian conjugate to eliminate terms proportional to $\mathbf{G}$ to obtain

$$\delta W = \frac{1}{4\mu_0} \int_0^1 d\psi [\Xi'^\dagger (\mathbf{F}\Xi' + \mathbf{K}\Xi) + \Xi^\dagger (\mathbf{F}\Xi' + \mathbf{K}\Xi)' + (\Xi'^\dagger \mathbf{F} + \Xi^\dagger \mathbf{K}^\dagger) \Xi' + (\Xi'^\dagger \mathbf{F} + \Xi^\dagger \mathbf{K}^\dagger)' \Xi]. \quad (64)$$

The integrand is an exact differential, allowing us to express the integral in terms of the values of the integrand at the end points (as in Newcomb's use of the Hilbert Invariant Integral)

$$\delta W = \frac{1}{4\mu_0} \int_0^1 d\psi [\Xi'^\dagger (\mathbf{F}\Xi' + \mathbf{K}\Xi) + (\Xi'^\dagger \mathbf{F} + \Xi^\dagger \mathbf{K}^\dagger) \Xi]' = \frac{1}{4\mu_0} [\Xi'^\dagger (\mathbf{F}\Xi' + \mathbf{K}\Xi) + (\Xi'^\dagger \mathbf{F} + \Xi^\dagger \mathbf{K}^\dagger) \Xi] \Big|_0^1. \quad (65)$$

For solutions that are regular at the magnetic axis, $\Xi(0) = 0$ and $(\mathbf{F}\Xi' + \mathbf{K}\Xi)(1) = \mathbf{W}(1)\Xi(1)$, where $\mathbf{W}$ is given by Eq. (55), which is self-adjoint. Thus

$$\delta W = \frac{1}{2\mu_0} \Xi^\dagger(1) \mathbf{W}_P \Xi(1), \quad \mathbf{W}_P \equiv \mathbf{W}(1). \quad (66)$$

VI. NUMERICAL METHODS AND RESULTS

The procedure developed above, called the Direct Method of Newcomb, has been implemented in the DCON code. In this section, we describe numerical methods and results.

DCON has been developed to treat any static equilibrium with isotropic fluid pressure which is constant on a flux surface, a solution to the Grad-Shafranov equation, Eq. (3). There are two methods for solving this equation,²⁵ Chapter 4: a direct method, which provides the solution $\chi(R, Z)$ on a rectangular domain in cylindrical coordinates $R_{\min} \leq R \leq R_{\max}$, $Z_{\min} \leq Z \leq Z_{\max}$; and an inverse method which provides a solution $R(\psi, \theta)$, $Z(\psi, \theta)$ on a domain of flux coordinates $0 < \psi < 1$, $0 \leq \theta < 1$. DCON reads either type of data, fits it to bicubic splines, and transforms it to an internal inverse representation in straight-fieldline coordinates, with a Jacobian $\mathcal{J}$ and bicubic spline nodes specified by the user. There is a collection of 28 interfaces to specific equilibrium solvers; additional interfaces are easy to implement. The bicubic spline representation enables smooth interpolation to any position (ψ, θ) . Equilibrium data are then used to compute the metric tensor components g_{ij} , Eq. (A1), as a function of ψ and θ , which are then Fourier analyzed using Eq. (14), for a range of M poloidal harmonics $m_{\text{low}} \leq m_{\text{high}}$. These are then used to compute the Hermitian matrices $\mathbf{F}$ and $\mathbf{G}$ and the non-Hermitian matrix $\mathbf{K}$ as functions of (ψ, m, m') , which are stored as complex cubic splines, using the same ψ nodes as the equilibrium data. Extensive tabular and graphic data are provided to understand and interpret the equilibrium. The internal inverse representation is used to compute and diagnose the relative error in the solution of the Grad-Shafranov equation. DCON computes and diagnoses the Mercier and ideal MHD infinite- n ballooning criteria.

The next step is to solve the Euler-Lagrange equation, Eq. (56), for the M column vectors of the solution matrix $\mathbf{U}$ which are regular at the magnetic axis. As discussed following Eq. (51), the magnetic axis is a singular point with M regular solutions. The initial conditions could in principle be obtained as a convergent power series, as is done for the singular series where $m = nq$. However, there is always numerical noise in the neighborhood of the axis because of inaccuracies in the Grad-Shafranov solution and the bicubic spline fits. Instead, we choose a regular point close to the axis, typically $\psi_{\text{low}} = 10^{-4}$, and initialize with $\mathbf{U}_{21} = 0$, $\mathbf{U}_{22} = \mathbf{I}$. Newcomb shows that this is equivalent in the limit $\psi_{\text{low}} \rightarrow 0$, and we find that this provides clean results. The differential equations are then integrated with adaptive integrator LSODE,⁹ which automatically adjusts its steps depending on how rapidly the solution varies. This is especially useful in the neighborhood of the axis, the singular surfaces, and the separatrix.

At each step of the integrator, we compute and monitor the critical determinant D_C , Eq. (57). A zero of D_C is the generalization of Newcomb's crossing criterion and indicates the presence of a fixed-boundary instability, as discussed in Section IV.

Our procedure requires that the column vectors of the solution matrix $\mathbf{U}$ retain their linear independence as the integration proceeds to increasing ψ . This is challenging because different independent solution vectors grow at greatly different rates with increasing ψ , and unavoidable numerical error can cause the smaller solution vectors to become polluted and eventually overwhelmed by the faster ones, resulting in a loss of linear independence. This causes the determinant to be dominated by noise and thus causes the

procedure to fail. This happens, for example, near the magnetic axis, where different independent solutions grow as $\psi^{m/2}$ for a large range of m , and can also happen near the singular surfaces. To avoid this, we have developed a method called Gaussian Reduction, also known informally as a fixup, similar to the Gaussian elimination used to factor matrices. We monitor the growth of the Euclidean norm of each solution vector and sort the solutions in decreasing order of this growth factor. When the ratio of the largest to the smallest growth factors exceeds a specified threshold, we stop the integration and apply the fixup. We subtract a linear multiple of the largest vector from each of the smaller vectors, so as to eliminate the largest component of the largest vector. Then, we repeat that operation for the next largest vector, and so on until the solution vector is triangularized. Numerical results show that this procedure leaves the determinant D_C unchanged and that its continued evolution remains smooth. In the absence of fixups, D_C eventually develops large noise, indicating a failure of the numerical method.

As the integration approaches each singular surface, it is terminated at a specified value of $|m - nq|$ for the singular value of m , and boundary conditions are applied to cross the singular surface, derived from the requirement that the solution be an element of the space of admissible functions with $\delta W < \infty$. Gaussian Reduction allows us to identify and eliminate the large resonant solution, reducing the multiplicity of solutions from M to $M - 1$. We restore the multiplicity to M by introducing a new component of the small resonant solution on the other side of the singular surface. Boundary conditions imposed on the nonresonant solutions are that their values and first derivatives be continuous, i.e., C^1 continuous. This procedure leaves D_C C^1 continuous. Using these conditions, the integration is continued an equal distance past the singular surface.

Wherever there is a zero of D_C , DCON reports the crossing and its location. Upon reaching the plasma-vacuum interface at $\psi = 1$, if there has been no change of the sign of D_C , DCON reports stability to fixed-boundary modes.

If the plasma region is bounded by a separatrix, the safety factor $q(\psi) \rightarrow \infty$ as $\psi \rightarrow 1$, and the integration must stop short of this singular point by a small amount, typically $\psi = 0.99$. If the $f'(\psi)$ and $P'(\psi)$ profiles in the Grad-Shafranov equation, Eq. (3), vanish at the plasma edge, then the current density vanishes there, and there is little error introduced by cutting off the plasma region there and treating it as vacuum. If the current density is finite at the edge and then drops suddenly to zero in the vacuum, that can excite edge-localized modes.

DCON then computes the plasma response matrix $\mathbf{W}$, Eq. (66), calls the VACUUM code²² to compute the vacuum response matrix $\mathbf{W}_V$, adds them to form the total response matrix $\mathbf{W}_T$, Eq. (62), and uses LAPACK routine ZHEEV to find and report its eigenvalues and eigenvectors in order of decreasing eigenvalue. The order of the complex matrix $\mathbf{W}_T$ is M , the number of Fourier components retained, which is generally small enough that the cpu time for this procedure is negligible. If $\mathbf{W}_T$ has any negative eigenvalues, the equilibrium is unstable to a free-boundary instability. Since $\mathbf{W}_V$

is positive-definite, a negative $\mathbf{W}_P$ must be the source of instability. Comparison of the plasma, vacuum, and total values enables an estimate of how close the mode is to marginal stability. In addition to the total eigenvalues, DCON also reports the contributions to the total due to the plasma and vacuum region. Eigenvectors of $\mathbf{W}_T$ are reported by DCON as a list of complex Fourier amplitudes, with the largest component indicated with an asterisk (*). For the smallest eigenvalue, this usually occurs for the m value closest to nq_a , where q_a is the safety factor at the edge.

As in any Fourier-based computation, the largest and smallest m components are inaccurate due to aliasing. This is mitigated by centering the spectrum about $m = nq_a$, and discounting the amplitudes reported at the ends of the spectrum and the stablest eigenvalues. Comparison between the largest and smallest Fourier amplitudes for the lowest eigenvalue gives an indication of how well-converged the smallest eigenvalues are. Convergence can be improved by adding additional components at the ends of the spectrum, at the expense of longer run time.

Since this procedure is based on minimizing only the potential energy δW , rather than the Lagrangian $\delta W - \omega^2 K$, the solutions are not eigenvectors of the motion, but only Euler-Lagrange test functions. They can provide insight into the motion of the plasma near marginal stability, where ω^2 is small. As discussed following Eq. (17), minimization of δW

leads immediately to considering only incompressible perturbations, $\nabla \cdot \xi = 0$. This decouples the system from the continuous spectrum and greatly improves the speed and accuracy of identifying the stability boundary, which is the real goal of this approach.

We now present an example of the use of DCON to determine the ideal MHD stability of an axisymmetric toroidal plasma a moderately high- β model of NSTX with $\beta_{\text{normal}} = 5.671$, with Grad-Shafranov solution computed by the EFIT code.²⁶ This run took 4.3 s on one core of a 2.5 GHz Intel Core i7 processor, using 128 radial and poloidal bicubic spline nodes for equilibrium representation and a relative tolerance for the LSODE adaptive integration of 10^{-6} . Stability results are given for toroidal mode number $n = 1$ and a range of poloidal harmonics $-12 \leq m \leq 26$, with the total number of harmonics $M = 39$. Figure 1 shows surface of constant psi and surfaces of constant Hamada coordinate θ .

Figure 2 shows profiles of scaled pressure $\mu_0 p$ and safety factor q vs. normalized radial coordinate ψ . The plasma region is bounded by a separatrix where $q \rightarrow \infty$, cut off at $\psi = 0.99$.

Figure 3 shows the two ways of visualizing the critical determinant D_C , Eq. (57) vs. ψ . This is the generalization of Newcomb's crossing criterion. Because it is the product of M eigenvalues, it often exceeds the floating point limit of

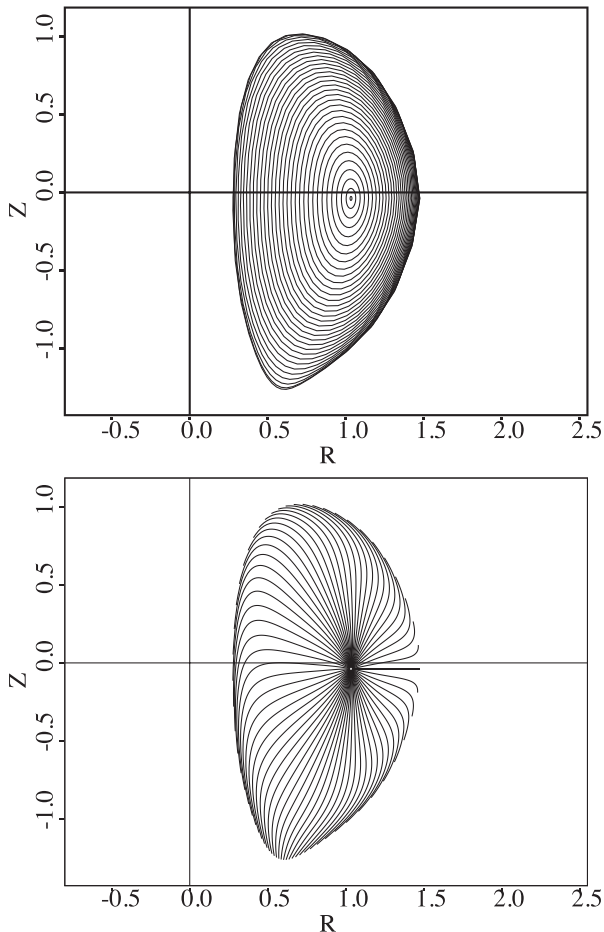


FIG. 1. NSTX equilibrium illustrating the use of ideal DCON. Top: poloidal magnetic flux surfaces. Bottom: lines of constant Hamada poloidal coordinate.

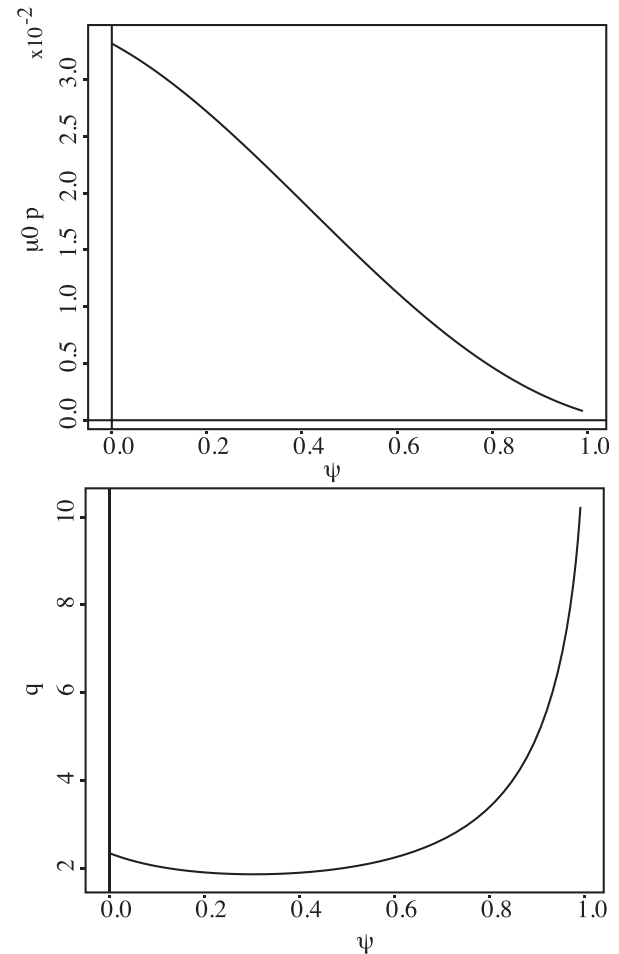


FIG. 2. Profiles for NSTX equilibrium. Top: scaled pressure $\mu_0 p$ vs. scaled flux coordinate ψ . Bottom: safety factor q vs. ψ .

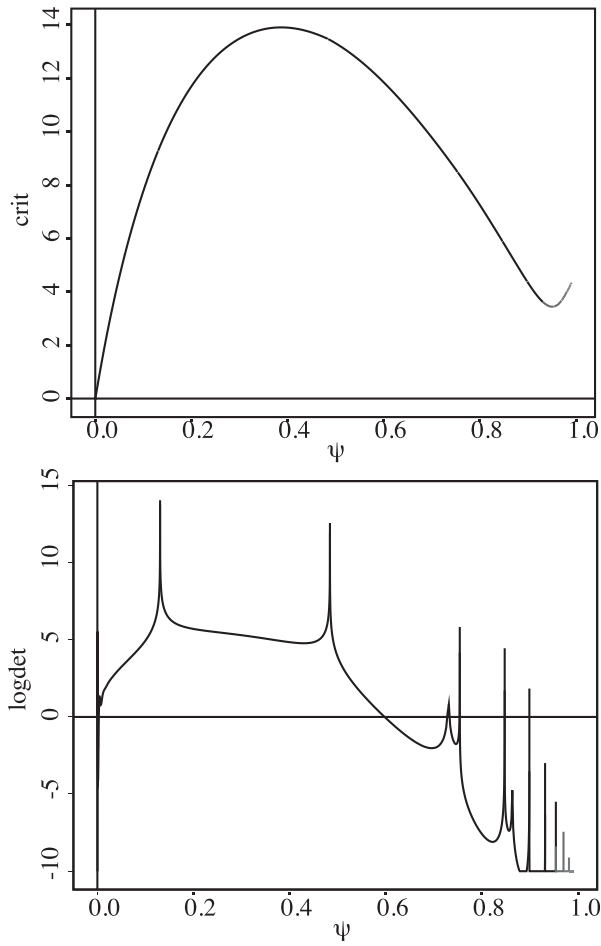


FIG. 3. The generalized Newcomb crossing criteria, described in the text. Top: crit, the smallest eigenvalue of $\mathbf{W}^{-1}$, vs. ψ . Bottom: logdet, the log of $|D_C|$, vs. ψ . These are for the original q profile with $q_0 > 1$.

Fortran, making it difficult to compute and graph. We have developed two ways of visualizing it. The first is to compute the eigenvalues of the $\mathbf{W}(\psi)$ matrix and graphing the smallest eigenvalue, which must change sign at a Newcomb crossing. This is shown on the left side of Fig. 3. In this case, there is no sign change and no fixed-boundary instability. A second method is to graph $\ln|D_C|$, as shown on the right of Fig. 3. This graph shows multiple upward-facing cusps, where $|D_C| \rightarrow \infty$, which have no physical significance. Cusps appear because of the finite step sizes taken by LSODE, which approach but do not reach the pole. There are no downward-facing cusps, which indicate that the equilibrium is stable.

In order to show the behavior of DCON for a case with a fixed-boundary instability, we modify the q -profile by adding a constant to f_0^2 on the right-hand side of the Grad-Shafranov equation, shifting $q(0)$ to 0.9, since $q < 1$ is known to be unstable. The results are shown in Fig. 4.

If the equilibrium is stable to fixed-boundary modes, the final step is to compute the plasma response matrix $\mathbf{W}_P$, call the VACUUM code to compute the vacuum response matrix $\mathbf{W}_V$, for the total response matrix $\mathbf{W}_T = \mathbf{W}_P + \mathbf{W}_V$, and compute and print its eigenvalues and eigenvectors in order of increasing eigenvalues. The most essential results are output to the screen, while full results are output to text file dcon.out. Extensive graphic diagnostics are also produced.

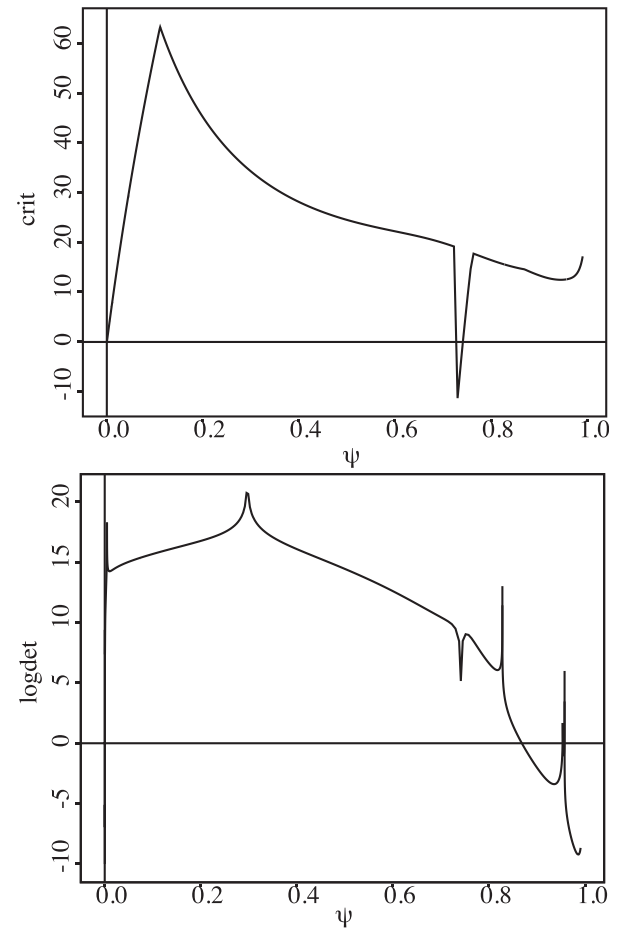


FIG. 4. The generalized Newcomb crossing criteria, described in the text. Top: crit, the smallest eigenvalue of $\mathbf{W}^{-1}$, vs. ψ . Bottom: logdet, the log of $|D_C|$, vs. ψ . These are for the modified q profile with $q_0 < 1$.

VII. DISCUSSION AND CONCLUSIONS

We have presented a method for determining the ideal MHD stability of axisymmetric toroidal plasmas to nonaxisymmetric perturbations. We have generalized the Direct Criterion of Newcomb (DCON) for the stability of fixed-boundary modes from his original cylindrical treatment to a full toroidal treatment. We have further extended the treatment to apply to free-boundary modes, with a moving boundary and a vacuum region.

The overall procedure involves a sequence of criteria. First, we test for Mercier stability, Eq. (49), on all flux surfaces. Any mode with a singular surface in a region where $D_I > 0$ is unstable. We then compute ballooning stability for ideal MHD modes in the limit as $n \rightarrow \infty$. If the equilibrium is Mercier and ballooning stable, we then test for stability to perturbations of each toroidal mode number $n \neq 0$. We integrate the Euler-Lagrange equations from the axis to the plasma-vacuum interface, initializing it at the axis and crossing each resonant surface according to the procedures discussed in Section IV. We monitor the critical determinant D_C (Eq. (57)). If it has a zero in the plasma region, then there is an ideal fixed-boundary instability, regardless of conditions at or beyond the plasma-vacuum interface. If there are no zeroes of D_C , we compute the plasma and

vacuum contributions to the potential energy matrix described in Section V and determine whether it has any negative eigenvalues. If so, there are free-boundary ideal instabilities. Otherwise, the system is stable to all ideal MHD perturbations.

It would be difficult to attempt a further generalization of this approach to nonaxisymmetric toroidal systems such as stellarators and torsatrons, as attempted by Bineau.¹⁰ For an axisymmetric system, in which the toroidal mode number n is a good quantum number, the resonant surfaces at $q = m/n$ are discrete and well-separated, except in unusual cases in which the axis is also a resonant surface or the magnetic shear q' vanishes at a rational surface. In a nonaxisymmetric system, since there are in principle infinitely many toroidal and poloidal harmonics, the singular surfaces would be dense everywhere, and this would appear to make the procedure impractical. Perhaps it could be shown that convergence is achieved for a moderate number of toroidal harmonics. Even then, the situation would be further complicated by the existence of stochastic regions of the equilibrium, especially in the neighborhood of low-order resonant surfaces, as occur in most nonaxisymmetric configurations, where flux surfaces, and hence flux coordinates, fail to exist.

A different sort of generalization is possible. In ideal MHD, at the resonant surfaces where $q = m/n$, the large resonant solution must be excluded in order to make the potential energy δW finite. This is a consequence of the well-known ideal MHD constraint that the field and the fluid move together. In the presence of a small resistivity, there is a narrow region where inertia and resistivity allow the field and the fluid to slip through each other, introducing a broad new class of resistive instabilities, or singular modes.^{13,14,19,27} These can be treated by the method of matched asymptotic expansions, partitioning the domain into stationary outer ideal MHD regions and inner regions including inertia and resistivity and other dynamic effects. The outer region satisfies the same equations as Eq. (21). Attempts to generalize the solution procedure presented here reveal that it is numerically unstable as a form of shooting method for a 2-point boundary value problem rather than an initial value method such as presented here. A generalization of the singular Galerkin method introduced in Refs. 13 and 14 will be presented in subsequent paper.

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APPENDIX: COEFFICIENT MATRICES

Here, we give detailed expressions for the coefficient matrices **A**, **B**, **C**, **D**, **E**, **H**, **F**, **G**, and **K**.

Before expressing these principal matrices, we define some more fundamental matrices in which are used in their definition. We define the following scalar components of the metric tensor:

$$\begin{aligned} g_{11} &\equiv |\nabla\theta \times \nabla\zeta|^2, \\ g_{22} &\equiv |\nabla\zeta \times \nabla\psi|^2, \\ g_{33} &\equiv |\nabla\psi \times \nabla\theta|^2, \\ g_{23} &\equiv (\nabla\zeta \times \nabla\psi) \cdot (\nabla\psi \times \nabla\theta), \\ g_{31} &\equiv (\nabla\psi \times \nabla\theta) \cdot (\nabla\theta \times \nabla\zeta), \\ g_{12} &\equiv (\nabla\theta \times \nabla\zeta) \cdot (\nabla\zeta \times \nabla\psi). \end{aligned} \quad (\text{A1})$$

For any scalar $f(\theta)$, we define an associated matrix $\langle f \rangle$ of Fourier coefficients

$$\langle f \rangle_{mm'} \equiv \int_0^1 d\theta \mathcal{J} \exp[2\pi i(m' - m)\theta] f(\theta). \quad (\text{A2})$$

Note that if f is real, then $\langle f \rangle$ is Hermitian, whereas if f is imaginary, then $\langle f \rangle$ is anti-Hermitian. The following quantities are required when the Jacobian is not uniform

$$J_{mm'} \equiv \langle 1 \rangle_{mm'}, \quad J'_{mm'} \equiv \langle \mathcal{J}' / \mathcal{J} \rangle_{mm'}. \quad (\text{A3})$$

Next, we define the matrices

$$(G_{ij})_{mm'} \equiv \langle g_{ij} \rangle_{mm'}. \quad (\text{A4})$$

Finally, we define the diagonal matrices **M** and **Q** in terms of their diagonal components

$$M_{m,m'} \equiv m\delta_{m,m'}, \quad Q_{m,m'} \equiv (m - nq)\delta_{m,m'}. \quad (\text{A5})$$

In terms of these fundamental matrices, the coefficient matrices appearing in Eq. (17) are given by

$$\begin{aligned} \mathbf{A} &= (2\pi)^2 [n(\mathbf{G}_{22} + \mathbf{G}_{23}\mathbf{M}) + \mathbf{M}(n\mathbf{G}_{23} + \mathbf{G}_{33}\mathbf{M})] \\ \mathbf{B} &= -2\pi i \chi' [n(\mathbf{G}_{22} + q\mathbf{G}_{23}) + \mathbf{M}(\mathbf{G}_{23} + q\mathbf{G}_{33})] \\ \mathbf{C} &= -2\pi i [\chi''(n\mathbf{G}_{22} + \mathbf{M}\mathbf{G}_{23}) \\ &\quad + (q\chi')'(n\mathbf{G}_{23} + \mathbf{M}\mathbf{G}_{33})] - (2\pi)^2 \chi' (n\mathbf{G}_{12} + \mathbf{M}\mathbf{G}_{31})\mathbf{Q} \\ &\quad + 2\pi i (2\pi f' \mathbf{Q} - nP' / \chi' \mathbf{J}) \\ \mathbf{D} &= \chi'^2 [(\mathbf{G}_{22} + q\mathbf{G}_{23}) + q(\mathbf{G}_{23} + q\mathbf{G}_{33})] \\ \mathbf{E} &= \chi' [\chi''(\mathbf{G}_{22} + q\mathbf{G}_{23}) + (q\chi')'(\mathbf{G}_{23} + q\mathbf{G}_{33}) \\ &\quad - 2\pi i \chi'^2 (\mathbf{G}_{12} + q\mathbf{G}_{31})\mathbf{Q} + P' \mathbf{J}] \\ \mathbf{H} &= \chi'' [\chi'' \mathbf{G}_{22} + (q\chi')' \mathbf{G}_{23}] + (q\chi')' [\chi'' \mathbf{G}_{23} + (q\chi')' \mathbf{G}_{33}] \\ &\quad + 2\pi i \chi' [\chi'' (\mathbf{M}\mathbf{G}_{12} - \mathbf{G}_{12}\mathbf{M}) + (q\chi')' (\mathbf{M}\mathbf{G}_{31} - \mathbf{G}_{31}\mathbf{M})] \\ &\quad + (2\pi \chi')^2 \mathbf{Q}\mathbf{G}_{11}\mathbf{Q} + [P'(\mathbf{J}\chi'' / \chi' + \mathbf{J}') - 2\pi f' q' \chi' \mathbf{I}]. \end{aligned} \quad (\text{A6})$$

Note also that the matrices **A**, **D**, and **H**, which couple like quantities in Eq. (17), are Hermitian, while **B**, **C**, and **E**, which couple unlike quantities, are not. The matrix **B** is anti-Hermitian because of the factor i .

Two of the composite matrices defined in Eq. (20) contain important cancellations. Thus, we can express

$$\begin{aligned}\mathbf{F} &\equiv \mathbf{D} - \mathbf{B}^\dagger \mathbf{A}^{-1} \mathbf{B} \\ &= (\chi'/n)^2 \mathbf{Q} [\mathbf{G}_{33} - (n\mathbf{G}_{23} + \mathbf{G}_{33}\mathbf{M}) \\ &\quad \times (2\pi)^2 \mathbf{A}^{-1} (n\mathbf{G}_{23} + \mathbf{M}\mathbf{G}_{33})] \mathbf{Q}.\end{aligned}\quad (\text{A7})$$

Note the presence of the factor $\mathbf{Q}$ on either side of $\mathbf{F}$, responsible for the vanishing of $\det \mathbf{F}$ at the resonant surfaces, discussed in detail in Section III. The remaining factor of $\mathbf{F}$, denoted $\bar{\mathbf{F}}$ in Eq. (29), is symmetric-positive-definite, which plays an essential role in Section IV.

Similarly, we can express

$$\begin{aligned}\mathbf{K} &= \mathbf{E} - \mathbf{B}^\dagger \mathbf{A}^{-1} \mathbf{C} \\ &= \mathbf{Q} (\chi'/n) \{ 2\pi i (n\mathbf{G}_{23} + \mathbf{G}_{33}\mathbf{M}) \mathbf{A}^{-1} \mathbf{C} \\ &\quad - [\chi'' \mathbf{G}_{23} + (q\chi')' \mathbf{G}_{33} - 2\pi i \chi' \mathbf{G}_{31} \mathbf{Q} + J_\theta \mathbf{I}] \}.\end{aligned}\quad (\text{A8})$$

No such simplification has been found for the remaining matrix $\mathbf{G}$ defined in Eq. (20).

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